

Yeonsu Chang

Integrated Ph.D Candidate

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Graph Theory

Structural Graph Theory

Parameterized Complexity

Reduced Parameter

Research Statement

My research focuses on structural aspects of graphs and related width parameters, with emphasis on characterizations and algorithmic consequences in parameterized complexity.

Education

Integrated Ph.D in Mathematics

Mar 2022 – Present

Hanyang University, Seoul, South Korea

Advisor: O-joung Kwon

B.S. in Mathematics

Mar 2015 – Feb 2022

Hanyang University, Seoul, South Korea (incl. two years of military service)

Military Service

Republic of Korea Army

Jan 2017 – Oct 2018

Completed mandatory military service (Sergeant)

Journal Papers

3. **Pre-assignment problem for unique minimum vertex cover on bounded clique-width graphs**
Shinwoo An, Yeonsu Chang, Kyungjin Cho, O-joung Kwon, Myounghwan Lee, Eunjin Oh, Hyeonjun Shin.
Theoretical Computer Science 1084 (Sep, 2026), 116173, *AAAI2025 accepted*, [arXiv:2408.09591](https://arxiv.org/abs/2408.09591)
2. **A new width parameter of graphs based on edge cuts: α -edge-crossing width**
Yeonsu Chang, O-joung Kwon, and Myounghwan Lee.
Discrete Applied Mathematics 380 (Feb, 2026), *WG23 accepted*, [arXiv:2302.04624](https://arxiv.org/abs/2302.04624)
1. **A characterization of graphs of radius- r flip-width at most 2**
Yeonsu Chang, Sejin Ko, O-joung Kwon, and Myounghwan Lee.
Discrete Mathematics 348(4) (Apr, 2025), 114366, [arXiv:2306.15206](https://arxiv.org/abs/2306.15206)

Refereed Conference Papers without Journal Version

1. **Moderately beyond clique-width: reduced component max-leaf and related parameters**
Édouard Bonnet, Yeonsu Chang, Julien Duron, Colin Geniet, and O-joung Kwon.
ESA 2026 accepted, [arXiv:2604.19138](https://arxiv.org/abs/2604.19138)

Preprints

1. **The optimal chromatic bound for even-hole-free graphs without induced seven-vertex paths**
Yeonsu Chang, Shenwei Huang, and Yidong Zhou.
[arXiv:2602.04403](https://arxiv.org/abs/2602.04403)

Talks

European Symposium on Algorithms 2026 (ESA 2026) <i>Moderately beyond clique-width: reduced component max-leaf and related parameters</i>	Aug 31– Sep 2, 2026
ASIACOMB 2026 <i>Algorithmic Applications of Reduced component max-leaf</i>	Aug 24– 28, 2026
Korean Mathematical Society 80th Anniversary Conference <i>The optimal chromatic bound for (even-hole, P_7)-free graphs</i>	Jun 22– 25, 2026
Korean Student Combinatorics Workshop 2026 Winter (KSCW2026W) <i>Structural and Algorithmic properties of Reduced Component Max-Leaf</i>	Feb 2– 6, 2026
2025 Korean Mathematical Society Annual Meeting <i>Structural and Algorithmic properties of Reduced Component Max-Leaf</i>	Oct 22– 24, 2025
Math Colloquium at Hanyang University <i>Combinatorial game and graph structure: cops and robber game and flipper game</i>	Nov 4, 2024
2023 KMS Annual Meeting <i>A characterization of graphs of radius-r flip-width at most 2</i>	Oct 26– 28, 2023

Teaching Experiences

Linear Algebra 1, Hanyang University	Spring 2025
Math Capstone PBL (Solving Graph Problems), Hanyang University	Fall 2024
Linear Algebra 1, Differential Geometry, Hanyang University	Spring 2024
Head TA, Hanyang University	Fall 2022 – Spring 2024
Math Capstone PBL (Applications of graph algorithms), Hanyang University	Fall 2022
Linear Algebra 1, Hanyang University	Spring 2022

Workshops and Conferences (In Korea)

2026 Summer School on Combinatorics and Algorithms KAIST, Daejeon, South Korea	Aug 10– 14, 2026
2026 Korean Student Combinatorics Workshop (KSCW 2026) Gongju, South Korea	Jul 27– 31, 2026
37th KIAS Combinatorics workshop Jeju, South Korea	May 28– 30, 2026
36th KIAS Combinatorics workshop KIAS, Seoul, South Korea	Mar 13– 14, 2026
2025 Korean Student Combinatorics Workshop Winter (KSCW 2025W) Yeosu, South Korea	Feb 2– 6, 2026
Workshop on Algorithm for Student 2026 POSTECH, Pohang, South Korea	Jan 26– 28, 2026
35th KIAS Combinatorics workshop	Dec 22– 24, 2025

Busan, South Korea	
2025 KMS Fall Meeting	Oct 22– 24, 2025
Seoul, South Korea	
2025 Korean Student Combinatorics Workshop (KSCW 2025)	Aug 20– 24, 2025
Gyeongju, South Korea	
2025 Combinatorics Workshop	Aug 18– 20, 2025
IBS, Daejeon, South Korea	
2025 Summer School on Combinatorics and Algorithms	Jul 14– 17, 2025
POSTECH, Pohang, South Korea	
34th KIAS Combinatorics workshop	May 29– 31, 2025
Jeju, South Korea	
2025 KMS Spring Meeting	Apr 24– 26, 2025
KAIST, Daejeon, South Korea	
IBS-DIMAG Winter School on Graph Minors	Feb 3– 19, 2025
IBS, Daejeon, South Korea	
One Day Workshop on Graph Theory	Jan 15, 2025
Ajou University, Suwon, South Korea	
The 4th Korean Logic Day	Jan 13– 15, 2025
Changwon, South Korea	
32th KIAS Combinatorics Workshop	Dec 19– 21, 2024
Busan, South Korea	
2024 KMS Fall Meeting	Oct 24– 26, 2024
Sungkyunkwan University, Suwon, South Korea	
2024 Korean Student Combinatorics Workshop (KSCW 2024)	Jul 29– Aug 2, 2024
Gongju, South Korea	
2024 Summer School on Combinatorics and Algorithms	Jul 22– 26, 2024
KIAS, Daejeon, South Korea	
Small workshop on parameterized complexity	Apr 30– May 1, 2024
POSTECH, Pohang, South Korea	
30th KIAS Combinatorics workshop	Mar 8– 9, 2024
KIAS, Seoul, South Korea	
29th KIAS Combinatorics workshop	Dec 26– 28, 2023
Busan, South Korea	
2023 KMS Fall Meeting	Oct 26– 28, 2023
Seoul National University, Seoul, South Korea	
2023 Combinatorics Workshop	Aug 16– 18, 2023
Yonsei University, Seoul, South Korea	
2023 KIAS Summer Workshop on Graph Theory	Aug 8– 9, 2023

KIAS, Seoul, South Korea

Lecture Series on Combinatorics at Hanyang University

Aug 1– 2, 2023

Hanyang University, Seoul, South Korea

28th KIAS Combinatorics workshop

Jun 1– 3, 2023

Jeju, South Korea

Seminar on Combinatorial Algorithms in Hanyang University

Apr 4– 5, 2023

Hanyang University, Seoul, South Korea

27th KIAS Combinatorics workshop

Feb 23– 24, 2023

KIAS, Seoul, South Korea

Workshop on Graph Coloring and Related Topics

Jan 28– 29, 2023

Konkuk University, Seoul, South Korea

26th KIAS Combinatorics workshop

Dec 20– 22, 2022

Busan, South Korea

2022 combinatorics workshop

Sep 30– Oct 1, 2022

GIST, Gwangju, South Korea

2022 School on Structural Graph Theory in Hanyang University

Sep 16– 17, 2022

Hanyang University, Seoul, South Korea

Combinatorics Incheon Workshop on Jeju

Jul 25– 29, 2022

Jeju National University, Jeju, South Korea

25th KIAS Combinatorics workshop

Jun 15– 18, 2022

Jeju, South Korea

Workshops and Conferences (International)

ALGO 2026

Aug 31– Sep 4, 2026

L'Aquila, Italy

ASIACOMB 2026

Aug 24– 28, 2026

Daejeon, South Korea

International Conference for the 80th Anniversary of the KMS

Jun 22– 25, 2026

Seoul, South Korea

Focused Workshop on Erdős–Pósa Problems

Mar 15– 20, 2026

Będlewo, Poland

SIAM Conference on Discrete Mathematics (DM24)

Jul 8– 11, 2024

Spokane, Washington, U.S.

3rd Korea-China Young Scholar's Symposium on Discrete Mathematics

Jun 24– 27, 2024

Jeju, South Korea

WG 2023 – Graph-Theoretic Concepts in Computer Science

Jun 28– 30, 2023

University of Fribourg, Fribourg, Switzerland

Research Visits

Academic Services

- Co-organizing 2026 Korean Student Combinatorics Workshop (KSCW 2026, 2026 조합론 학생 워크샵)

Awards

Best Paper Award. "A characterization of graphs of radius- r flip-width at most 2" Feb, 2026
Graduate Student Paper Competition, Institute of Natural Sciences, Hanyang University